IGCSE 0607 Practice Questions — E1.17

Surds

Total Questions: 39 — **Total Marks:** 191 Marks

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- **E1.17** Surds
  → **Paper 2 (Non-Calculator)** *(39 Qs)*

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1 Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

2 Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

3 Simplify fully.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

4 Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

5 Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

6 **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

(b) Simplify.

$\sqrt{20} + \sqrt{80}$

..... **[2]**

7 Simplify

$$\sqrt{75} + \sqrt{27} - \sqrt{12}$$

..... **[3]**

8 Simplify the following expression, leaving your answer in the form $k\sqrt{3}$.

$\sqrt{48} - \sqrt{12}$

..... **[3]**

9 **(a)** Simplify.

$$\sqrt{75} + \sqrt{12}$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

10 **(a)** Simplify

$$\sqrt{75} - \sqrt{12}$$

..... **[2]**

(b) Rationalise the denominator and simplify

$$\frac{18}{\sqrt{6}}$$

..... **[2]**

11 **(a)** Simplify.

$\sqrt{75}$

. . . . . [**2**]

(b) Simplify.

$2\sqrt{18} - \sqrt{8}$

. . . . . [**2**]

12 **(a)** **(i)** Simplify.

$$\sqrt{45} + \sqrt{20}$$

..... **[2]**

(ii) Rationalise the denominator.

$$\frac{18}{\sqrt{3}}$$

..... **[2]**

13 **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

(b) Work out the value of the following expression, giving your answer in the form $k\sqrt{2}$.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

(c) Rationalise the denominator.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

14 **(a)** Expand and simplify.

$$(4 - \sqrt{3})(2 + \sqrt{3})$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

15 **(a)** Expand and simplify

$$(3 - \sqrt{5})^2$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

16 **(a)** Expand and simplify.

$(3 - 2\sqrt{5})(4 + \sqrt{5})$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

17 (a) (a) Expand and simplify

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{15}{\sqrt{3}}$$

..... **[2]**

18 **(a)** Expand and simplify

$$(3+\sqrt{2})(4-\sqrt{2})$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{22}{5-\sqrt{3}}$$

..... **[3]**

19 **(a)** Expand and simplify.

$(3 + \sqrt{2})^2$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{4}{3 - \sqrt{5}}$$

..... **[3]**

20 **(a)** Expand and simplify.

$$(2\sqrt{5}-3)(2\sqrt{5}+3)$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{3-\sqrt{2}}$$

..... **[3]**

21 **(a)** Expand and simplify.

$$(3\sqrt{2}-1)(2\sqrt{2}+5)$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{26}{4-\sqrt{3}}$$

..... **[3]**

22 **(a)** Expand and simplify

$$(3-\sqrt{5})^2$$

..... **[2]**

(b) Write $\dfrac{11}{4-\sqrt{5}}$ in the form $a+b\sqrt{5}$, where a and b are integers.

..... **[3]**

23 **(a)** Rationalise the denominator and simplify fully.

$$\frac{15}{\sqrt{5}}$$

..... **[2]**

(b) Expand and simplify.

$$(2+\sqrt{3})(5-\sqrt{3})$$

..... **[3]**

24 **(a)** Expand and simplify.

$$(3+\sqrt{5})(2-\sqrt{5})$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[3]**

25 **(a)** Expand and simplify.

$$(5 - 2\sqrt{7})(3 + \sqrt{7})$$

..... **[2]**

(b) Rationalise the denominator and simplify fully.

$$\frac{22}{5 - \sqrt{3}}$$

..... **[3]**

26 **(a)** Expand and simplify.

$(4 - 3\sqrt{2})(5 + 2\sqrt{2})$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{11}{4 - \sqrt{5}}$$

..... **[3]**

(c) Solve the equation.

$x\sqrt{27} - 8 = x\sqrt{3} + 10$

Give your answer in the form $k\sqrt{3}$ where k is an integer.

$x =$ ..... **[3]**

27 A rectangle has an area of $5 + \sqrt{7}$ cm^2 and a width of $3 - \sqrt{7}$ cm.
Calculate the length of the rectangle.
Give your answer in the form $a + b\sqrt{7}$ where a and b are integers.

..... *cm* **[4]**

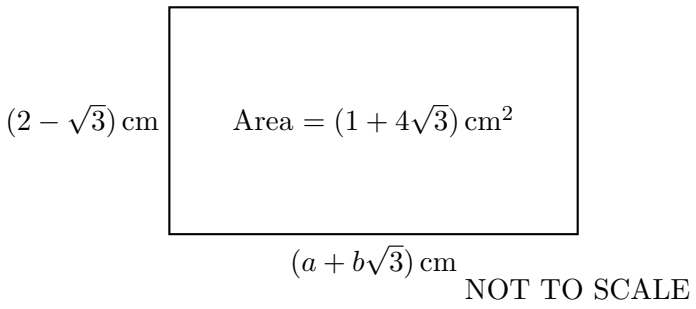
28 Solve the equation.

$$x\sqrt{3} - 2 = x + \sqrt{12}$$

Give your answer in the form $a + b\sqrt{3}$, where a and b are integers.

$x =$ ..... **[5]**

29 The diagram shows a rectangle with width $(2 - \sqrt{3})$ cm and length $(a + b\sqrt{3})$ cm, where a and b are integers.
The area of the rectangle is $(1 + 4\sqrt{3})\,\text{cm}^2$.



NOT TO SCALE

Work out the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[5]**

30 **(a)** Rationalise the denominator and simplify.

$$\frac{5+\sqrt{3}}{2-\sqrt{3}}$$

..... **[3]**

(b) Solve the equation.
Give your answer in the form $a+b\sqrt{2}$, where a and b are integers.

$x(3-\sqrt{2})=14$

$x =$ ..... **[3]**

31 **(a)** Express $\dfrac{5+\sqrt{3}}{2-\sqrt{3}}$ in the form $a+b\sqrt{3}$, where a and b are integers.

..... **[3]**

(b) A rectangle has an area of $12+4\sqrt{5}$ square centimetres and a width of $3+\sqrt{5}$ centimetres. Calculate the exact length of the rectangle, giving your answer in its simplest form.

Length = ... cm **[4]**

32 **(a)** Rationalise the denominator and simplify.

$$\frac{22}{5-\sqrt{3}}$$

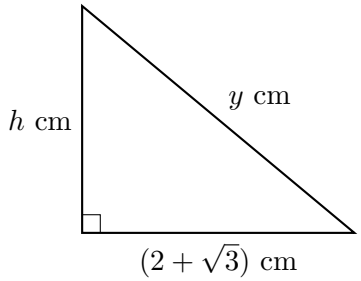
..... **[3]**

(b) Solve the equation.

$x\sqrt{2} = x + \sqrt{8}$ Give your answer in the form $a + b\sqrt{2}$, where a and b are integers.

$x =$ ..... **[4]**

33 A right-angled triangle has a base of $(2 + \sqrt{3})$ cm and an area of $(5 + 3\sqrt{3})$ cm^2.



(a) Show that the exact height, h, is $(2 + 2\sqrt{3})$ cm.

[4]

(b) The hypotenuse of the triangle is y cm.
Work out the exact value of y^2.
Give your answer in the form $a + b\sqrt{3}$, where a and b are integers.

$y^2 =$ ..... **[3]**

34 **(a)** A rectangle has an area of $(5 + 2\sqrt{3})$ cm^2 and a length of $(2 + \sqrt{3})$ cm.

Work out the width of the rectangle in the form $a + b\sqrt{3}$, where a and b are integers.

..... cm **[3]**

(b) Solve the equation.

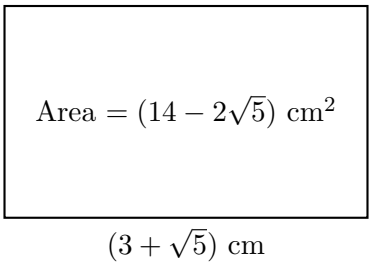
$$x\sqrt{5} - 2 = 2x + \sqrt{20}$$

Give your answer in the form $c + d\sqrt{5}$, where c and d are integers.

Show all your working.

$x =$ ..... **[4]**

35 A rectangle has length $(3 + \sqrt{5})$ cm and area $(14 - 2\sqrt{5})$ cm^2.



(a) Calculate the width of the rectangle.
Give your answer in the form $a + b\sqrt{5}$, where a and b are integers.

..... cm **[4]**

(b) The length of the diagonal of the rectangle is d cm.
Show that $d^2 = 308 - 124\sqrt{5}$.

[3]

36 **(a)** Rationalise the denominator and simplify fully.

$$\frac{6}{3-\sqrt{3}}$$

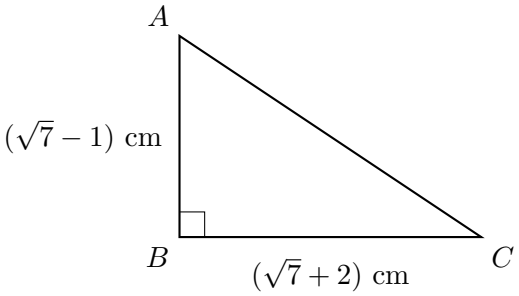
..... **[4]**

(b) Write the following expression in the form $a\sqrt{2}+b\sqrt{3}$, where a and b are integers.

$$\frac{\sqrt{24}+\sqrt{54}}{\sqrt{3}}-\frac{10}{\sqrt{2}}$$

..... **[3]**

37 The diagram shows a right-angled triangle ABC.
$AB = (\sqrt{7} - 1)$ cm, $BC = (\sqrt{7} + 2)$ cm and angle $ABC = 90°$.



(a) Calculate the area of triangle ABC.
Give your answer in the form $\dfrac{p + q\sqrt{7}}{2}$, where p and q are integers.

Area = ... cm^2 **[3]**

(b) Find the length of AC in the form $\sqrt{a + b\sqrt{7}}$, where a and b are integers.

AC = ... cm **[4]**

38 A right-angled triangle has a base of length $(3 + \sqrt{2})$ cm and an area of $(5 + 2\sqrt{2})$ cm^2.

(a) Show that the height of the triangle is $\dfrac{2(5 + 2\sqrt{2})}{3 + \sqrt{2}}$ cm.

[2]

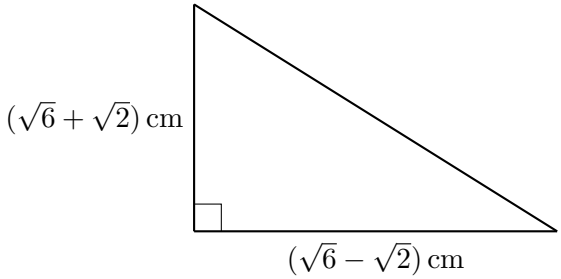
(b) Rationalise the denominator of $\dfrac{2(5 + 2\sqrt{2})}{3 + \sqrt{2}}$ to express the height in the form $\dfrac{a + b\sqrt{2}}{c}$, where a, b and c are integers.

..... cm **[4]**

(c) Given instead that another expression for a length is $\dfrac{1}{\sqrt{7} - 2}$, write this length in the form $\dfrac{\sqrt{7} + c}{d}$ where c and d are integers.

..... **[2]**

39 The diagram shows a right-angled triangle.



(a) Calculate the area of the triangle.

..... cm^2 **[3]**

(b) Show that the length of the hypotenuse is 4 cm.

[4]

(c) Another right-angled triangle is mathematically similar to this triangle.
The area of the similar triangle is 18 cm^2.
Calculate the length of its hypotenuse.

..... cm **[3]**